

Space Debris Removal

Background for Robotics & Controls Engineers

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Mission Concept for Mitigating Space Debris in LEO

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1 Space Debris Problem Overview

1.1 What is Space Debris?

- **Defunct satellites:** No longer operational spacecraft
- **Rocket bodies:** Upper stages left in orbit
- **Fragmentation debris:** Pieces from collisions and explosions
- **Mission-related objects:** Lens caps, separation mechanisms, etc.

1.2 LEO Characteristics

- **Altitude:** 200–2000 km above Earth
- **Orbital velocity:** ~ 7.8 km/s (28,000 km/h!)
- **Orbital period:** ~ 90 –120 minutes
- **Atmospheric drag:** Still present, causes gradual decay

1.3 Why It Matters

- Over 34,000 tracked objects > 10 cm
- ~ 1 million objects between 1–10 cm
- Collision at orbital speeds can destroy spacecraft
- Even small debris (1 cm) can be catastrophic

2 Orbital Mechanics Fundamentals

2.1 Kepler's Laws

As a controls engineer, you need to understand that satellites follow predictable paths:

2.1.1 Elliptical Orbits

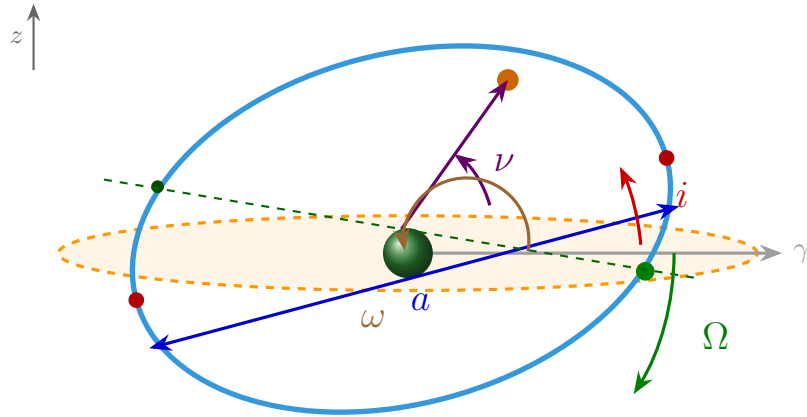
- Orbits are ellipses with Earth at one focus
- Described by semi-major axis (a) and eccentricity (e)

2.1.2 Orbital Parameters (State Vector)

Position: $[x, y, z]$
Velocity: $[v_x, v_y, v_z]$

Or using **Keplerian Elements**:

- a : Semi-major axis (orbit size)
- e : Eccentricity (orbit shape: 0 = circular, < 1 = ellipse)
- i : Inclination (tilt relative to equator)
- Ω : Right Ascension of Ascending Node (RAAN)
- ω : Argument of Periapsis
- ν : True Anomaly (position in orbit)

**Legend:**

- **Green sphere:** Earth (at focus of ellipse)
- **Orange dashed ellipse:** Equatorial plane (reference plane)
- **Blue ellipse:** Orbital plane (satellite's path)
- γ : Vernal equinox direction (reference for Ω)
- **Orange dot:** Satellite position
- **Red dots:** Periapsis (closest) and apoapsis (farthest) points
- **Green dots:** Ascending node (orbit crosses equatorial plane going north)

Six Keplerian Elements:

- a : **Semi-major axis** – size of orbit (average radius)
- e : **Eccentricity** – shape of orbit ($0 = \text{circle}$, $<1 = \text{ellipse}$); shown by ellipse elongation
- i : **Inclination** – tilt angle between orbital plane and equatorial plane
- Ω : **Right Ascension of Ascending Node (RAAN)** – angle from γ to ascending node
- ω : **Argument of periapsis** – angle from ascending node to periapsis
- ν : **True anomaly** – angle from periapsis to satellite (changes with time)

Figure 1: Keplerian Orbital Elements in 3D perspective. The six elements $(a, e, i, \Omega, \omega, \nu)$ uniquely define a satellite's position and velocity at any time.

2.2 Two-Body Problem

The fundamental equation of orbital motion:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} \quad (1)$$

Where:

- $\mu = GM$ (gravitational parameter of Earth = 398,600 km³/s²)
- $\mathbf{r}$ = position vector
- $\ddot{\mathbf{r}}$ = acceleration

Key Insight for Controls

In orbit, there's no friction! Any velocity change is permanent unless you apply another force.

2.3 Hohmann Transfer

To change orbits efficiently:

1. Apply Δv at periapsis to raise apoapsis
2. Coast to new apoapsis
3. Apply Δv to circularize

Critical for your problem: Rendezvous requires matching both position AND velocity!

3 The Kessler Syndrome

3.1 The Cascading Collision Problem

Named after NASA scientist Donald Kessler (1978), this is the critical threat:

1. Two objects collide at ~ 15 km/s relative velocity
2. Collision creates thousands of new debris fragments
3. Each fragment can cause more collisions
4. **Exponential growth** of debris population
5. Eventually, LEO becomes unusable (chain reaction)

3.2 Current Risk Level

- We're approaching the critical density threshold
- Some orbits (800–1000 km) are already at risk
- Mega-constellations (Starlink: 42,000 satellites planned) increase risk

3.3 Why Active Removal is Needed

- Natural decay takes decades-to-centuries in LEO
- Atmospheric drag only significant below ~ 600 km
- We need to remove ~ 5 –10 large objects per year to stabilize

4 Spacecraft Dynamics & Control in Space

4.1 Attitude Dynamics (Rotation)

Euler's Rotational Equation:

$$\mathbf{I} \cdot \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I} \cdot \boldsymbol{\omega}) = \boldsymbol{\tau} \quad (2)$$

Where:

- $\mathbf{I}$ = Moment of inertia tensor (3×3 matrix)
- $\boldsymbol{\omega}$ = Angular velocity vector
- $\boldsymbol{\tau}$ = Applied torque

Attitude Representation:

- Euler angles (intuitive but have singularities)
- Quaternions (4 parameters, no singularities – preferred!)
- Rotation matrices (9 parameters, redundant)

4.2 Actuators in Space

4.2.1 For Translation (Δv):

1. **Thrusters:** Chemical or electric propulsion

- Impulse: $\Delta v = v_e \cdot \ln(m_0/m_f)$ – Tsiolkovsky equation
- Fuel limited!

4.2.2 For Rotation (torque):

1. **Reaction Wheels:** Spin internal mass, conserve angular momentum

- Fast, precise, can saturate (need desaturation)

2. **Control Moment Gyroscopes (CMGs):** Gimbaled spinning wheels

- Higher torque than reaction wheels

3. **Magnetic Torquers:** Interact with Earth's magnetic field

- Slow, only certain directions, free propellant

4. **Thrusters:** Fast but uses fuel

4.3 Key Control Challenges

- **Underactuated system:** Limited fuel, must be efficient
- **Coupled dynamics:** Translation-rotation coupling during capture
- **Long time delays:** Ground communication ~1 second delay
- **Sensor noise:** Star trackers, gyros, GPS in space

5 Rendezvous and Proximity Operations (RPO)

5.1 The Four Phases of Rendezvous

Phase 1: Phasing (Far Range, >100 km)

- Match orbital plane (inclination)
- Adjust orbit to approach target
- Use ground-based tracking

Phase 2: Approach (10–100 km)

- Refine trajectory
- Begin autonomous sensing
- Switch to relative navigation

Phase 3: Proximity Operations (100 m – 10 km)

- **Critical phase for your problem!**
- Careful trajectory planning
- Collision avoidance
- Continuous target tracking

Phase 4: Close Proximity & Capture (<100 m)

- Final approach corridor
- Synchronize rotation rates
- Execute capture

5.2 Hill-Clohessy-Wiltshire (HCW) Equations

Relative motion in orbit (linearized):

$$\ddot{x} - 2n\dot{y} - 3n^2x = \frac{f_x}{m} \quad (3)$$

$$\ddot{y} + 2n\dot{x} = \frac{f_y}{m} \quad (4)$$

$$\ddot{z} + n^2z = \frac{f_z}{m} \quad (5)$$

Where:

- (x, y, z) : Local orbital frame (radial, along-track, cross-track)
- n : Mean motion (orbital angular velocity) = $\sqrt{\mu/a^3}$
- $\mathbf{f}$: Control forces

Key Insight

- Radial (x) and along-track (y) are **coupled** via Coriolis terms ($2n\dot{y}$, $2n\dot{x}$)
- Cross-track (z) is **decoupled**
- Natural motion creates elliptical relative trajectories

5.3 V-bar and R-bar Approaches

R-bar (Radial approach):

- Approach from above/below
- Naturally stable against along-track drift
- Requires active station-keeping

V-bar (Velocity vector approach):

- Approach from behind/front
- Passive safety (natural separation if engines fail)
- Preferred for ISS

6 Capture Mechanisms

6.1 Contact-Based Methods

6.1.1 1. Robotic Arm

- Example: Canadarm2 on ISS, SSRMS
- **Pros:** Precise, proven technology
- **Cons:** Requires close proximity, complex dynamics
- **Control Challenge:** Free-floating base problem!

Free-Floating Dynamics

When arm moves $\rightarrow$ spacecraft base reacts (Newton's 3rd law)
You need **Generalized Jacobian** that accounts for base motion:

- Position of end-effector depends on both joint angles AND base motion
- Captures induce momentum transfer $\rightarrow$ spacecraft tumbles

6.1.2 2. Harpoon

- Fire penetrating spike into target
- Tether connects chaser to target
- **Pros:** Works for non-cooperative targets
- **Cons:** Can fragment target, tether dynamics complex

6.1.3 3. Net

- Deploy net to envelop target
- **Pros:** Gentle, encapsulates fragments
- **Cons:** Deployment dynamics, tangling risk

6.1.4 4. Gripper/Grapple

- Mechanical jaws grip target
- Requires attachment point (cooperative targets)

6.2 Contactless Methods

6.2.1 1. Electrostatic Tractor

- Charge spacecraft oppositely → Coulomb attraction
- **Pros:** No contact, scalable force
- **Cons:** Requires charge management, works at close range

6.2.2 2. Laser Ablation

- Vaporize surface material → creates thrust on target
- **Pros:** Standoff distance, no rendezvous needed
- **Cons:** Debris from ablation, power requirements

6.2.3 3. Ion Beam Shepherd

- Plasma thruster creates force on target
- **Pros:** Gentle, continuous force
- **Cons:** Very slow, complex plasma dynamics

6.2.4 4. Magnetic Tractor

- Use magnetic fields (for ferromagnetic debris)
- **Pros:** No contact
- **Cons:** Only works for magnetic materials

7 Control Challenges for Non-Cooperative Targets

7.1 What Makes a Target “Non-Cooperative”?

1. **Unknown dynamics:** Tumbling rate unknown
2. **No communication:** Can't command it
3. **Unknown mass properties:** Center of mass, inertia tensor unknown
4. **Unknown shape/structure:** Visual only
5. **Unpredictable:** Solar radiation pressure, atmospheric drag variations

7.2 State Estimation Problem

You need to estimate:

- **Position & velocity** (6 DOF translation)
- **Orientation & angular velocity** (6 DOF rotation)
- **Total: 12-state estimation problem**

Sensors:

- **LIDAR:** Range and range-rate
- **Cameras:** Visual tracking, structure from motion
- **Radar:** All-weather tracking

Estimation Algorithms:

- Extended Kalman Filter (EKF)
- Unscented Kalman Filter (UKF) – better for nonlinear
- Particle Filters – for highly nonlinear/multimodal

7.3 Tumbling Dynamics

Debris often tumbles due to:

- Collision-induced rotation
- Gravity gradient torque
- Solar radiation pressure
- Magnetic field interaction

Euler's Equation (torque-free motion):

$$I_1 \dot{\omega}_1 = (I_2 - I_3) \omega_2 \omega_3 \quad (6)$$

$$I_2 \dot{\omega}_2 = (I_3 - I_1) \omega_3 \omega_1 \quad (7)$$

$$I_3 \dot{\omega}_3 = (I_1 - I_2) \omega_1 \omega_2 \quad (8)$$

Result: Complex 3D tumbling, periodic for rigid bodies

7.4 Detumbling Strategies

Option 1: Match Rotation

- Estimate target angular velocity
- Synchronize chaser rotation
- Capture in rotating frame
- **Challenge:** High-rate tumbling $\rightarrow$ high control effort

Option 2: Active Detumbling Before Capture

- Apply external torque (contactless methods)
- Slow down tumbling gradually
- Then capture
- **Challenge:** Takes time, requires precise torque control

Option 3: Capture While Tumbling

- Fast capture mechanism
- Accept the momentum transfer
- Detumble combined system afterward
- **Challenge:** Huge impact forces, risk of damage

8 Deorbiting Techniques

8.1 Natural Decay

- Atmospheric drag gradually lowers orbit
- Decay time = $f(\text{altitude, area-to-mass ratio, solar activity})$
- **Problem:** Can take decades above 600 km

8.2 Active Deorbiting

8.2.1 1. Direct Re-entry

- Use propulsion to lower periapsis below ~ 80 km
- Atmospheric drag increases exponentially
- Object burns up
- **Δv required:** $\sim 100\text{--}200$ m/s from LEO

8.2.2 2. Drag Augmentation

a) Drag Sail

- Deploy large surface area (sail)
- Increases drag force: $F_{\text{drag}} = 0.5\rho v^2 C_D A$
- Accelerates natural decay
- **Pros:** Passive, no propellant
- **Cons:** Attitude stabilization needed

b) Inflatable Structures

- Similar to drag sail but lighter
- Can increase area by $10\text{--}100\times$

c) Electrodynamic Tether

- Long conducting wire in Earth's magnetic field
- Generates current $\rightarrow$ Lorentz force $\rightarrow$ drag
- **Pros:** No propellant, can generate power
- **Cons:** Complex deployment, tether dynamics

8.2.3 3. Graveyard Orbit

- Boost to higher orbit (>2000 km)
- Out of useful LEO region
- **Problem:** Doesn't solve long-term issue

8.3 Controlled Re-entry

For large objects (>1 ton):

- Target specific ocean areas (South Pacific)
- Prevent casualty on ground
- Requires precise trajectory control

9 Key Differences from Terrestrial Robotics

1. No Fixed Base

- Manipulator motion $\rightarrow$ base reaction
- Need **Generalized Jacobian** and **Dynamic Singularities**
- Momentum conservation critical

2. No Gravity (Microgravity)

- Objects don't "fall"
- Small forces persist
- Contact forces dominate

3. Orbital Dynamics

- Reference frame accelerates (centripetal)
- Coriolis effects in relative motion
- Tidal forces (gravity gradient)

4. Propellant Constraints

- Every maneuver costs fuel
- Fuel mass = mission lifetime
- Optimization critical

5. Extreme Environment

- Temperature extremes: -150°C to $+150^{\circ}\text{C}$
- Radiation damages electronics

- Vacuum $\rightarrow$ no convective cooling
- Atomic oxygen erodes materials

6. Communication Delays

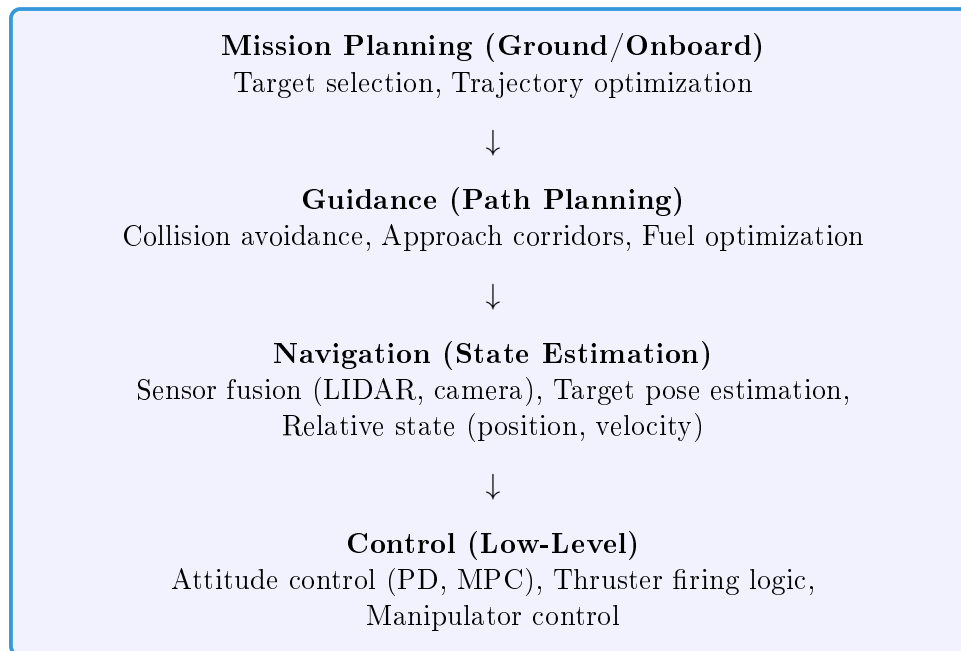
- Ground stations: ~ 1 sec delay
- Requires autonomy
- Can't teleoperate fast maneuvers

7. Sensing Challenges

- GPS limited to LEO
- Visual tracking affected by lighting (sun/eclipse)
- No tactile feedback before contact

10 Control Architecture for ADR Mission

10.1 Typical System Layers



10.2 Control Algorithms Relevant for This Problem

1. PID Control

- Simple, robust
- Works for attitude stabilization
- May struggle with coupled dynamics

2. Model Predictive Control (MPC)

- Optimal for constrained systems
- Handles thruster saturation
- Fuel optimization

- Collision avoidance constraints
- **Highly recommended for proximity ops!**

3. Sliding Mode Control

- Robust to uncertainties
- Good for non-cooperative targets
- Chattering can waste fuel

4. Adaptive Control

- Estimate unknown target parameters online
- Adjust control gains
- Useful for tumbling targets

5. Optimal Control (LQR/LQG)

- Minimize fuel + time
- Well-suited for rendezvous

11 Recommended Focus Areas for Your Solution

11.1 As a Robotics/Controls Engineer, You Can Contribute:

1. Robust State Estimation

- Develop filters for tumbling target tracking
- Sensor fusion (vision + LIDAR)
- Handle occlusions, lighting changes

2. Trajectory Planning

- Fuel-optimal approach trajectories
- Collision avoidance
- Reachability analysis

3. Capture Control

- Free-floating base control
- Impact minimization
- Post-capture stabilization

4. Detumbling Strategies

- Contactless detumbling (if using ion beam, etc.)
- Momentum management
- Combined system control after capture

5. Fault Tolerance

- Thruster failures
- Sensor degradation
- Safe-mode behaviors

12 Solved Examples: From Simple to Advanced

This section contains 6 worked examples to help you understand orbital mechanics and space debris removal concepts. Each example includes a detailed, step-by-step solution explained in simple terms.

12.1 Example 1: Calculating Orbital Velocity (EASY)

Problem

A satellite is orbiting Earth at an altitude of 400 km. What is its orbital velocity?

Given:

- Altitude: $h = 400$ km
- Earth's radius: $R_E = 6371$ km
- Gravitational parameter: $\mu = 398600 \text{ km}^3/\text{s}^2$

ELI10 Concept: Think of a satellite like a ball on a string that you're swinging around your head. The faster you swing it, the harder it pulls outward. In space, gravity is like that string pulling the satellite inward. The satellite needs to go just fast enough so that gravity's pull exactly balances its tendency to fly away. Too slow and it falls to Earth; too fast and it escapes into space!

Solution:

Step 1: Calculate orbital radius r (distance from Earth's center)

$$r = R_E + h = 6371 + 400 = 6771 \text{ km} \quad (9)$$

Why? The satellite isn't orbiting at Earth's surface but 400 km above it. We measure orbits from Earth's center, not the surface, so we add the radius.

Step 2: Use the orbital velocity formula

$$v = \sqrt{\frac{\mu}{r}} = \sqrt{\frac{398600}{6771}} = \sqrt{58.85} = 7.67 \text{ km/s} \quad (10)$$

Step 3: Convert to more familiar units

$$v = 7.67 \times 3600 = 27,612 \text{ km/h} \approx 27,600 \text{ km/h} \quad (11)$$

Answer

The satellite travels at **7.67 km/s** or about **27,600 km/h**! That's about 70 times faster than a commercial airplane. At this speed, the satellite completes one orbit around Earth in about 90 minutes.

Reality Check: The International Space Station orbits at a similar altitude and travels at approximately this speed. You can actually see it passing overhead in the evening!

12.2 Example 2: How Long Does Orbit Take? (EASY-MEDIUM)

Problem

Calculate the orbital period (time for one complete orbit) of a GPS satellite at an altitude of 20,200 km.

Given:

- Altitude: $h = 20,200$ km
- Earth's radius: $R_E = 6371$ km
- $\mu = 398600 \text{ km}^3/\text{s}^2$

ELI10 Concept: Imagine you're on a merry-go-round. If you stand close to the center, you go around quickly. If you stand at the edge, it takes longer to complete one rotation. Satellites work the same way – the farther from Earth, the longer one orbit takes!

Solution:

Step 1: Calculate orbital radius

$$r = R_E + h = 6371 + 20200 = 26,571 \text{ km} \quad (12)$$

Step 2: Calculate semi-major axis a (for circular orbit, $a = r$)

$$a = 26,571 \text{ km} \quad (13)$$

Step 3: Use Kepler's Third Law to find orbital period

$$T = 2\pi \sqrt{\frac{a^3}{\mu}} = 2\pi \sqrt{\frac{26571^3}{398600}} \quad (14)$$

Step 4: Calculate step by step

$$a^3 = 26571^3 = 1.876 \times 10^{13} \text{ km}^3 \quad (15)$$

$$\frac{a^3}{\mu} = \frac{1.876 \times 10^{13}}{398600} = 4.707 \times 10^7 \text{ s}^2 \quad (16)$$

$$\sqrt{\frac{a^3}{\mu}} = \sqrt{4.707 \times 10^7} = 6859 \text{ s} \quad (17)$$

$$T = 2\pi \times 6859 = 43,082 \text{ s} \quad (18)$$

Step 5: Convert to hours

$$T = \frac{43082}{3600} = 11.97 \text{ hours} \approx 12 \text{ hours} \quad (19)$$

Answer

The GPS satellite completes one orbit in **12 hours** (exactly 11 hours 58 minutes).

Cool Fact: GPS satellites are designed so that exactly 2 complete orbits = 1 Earth day (24 hours). This means they appear in the same position in the sky every day at the same time. That's why your GPS can always find at least 4 satellites overhead!

12.3 Example 3: Hohmann Transfer - Moving Between Orbits (MEDIUM)

Problem

A debris removal satellite needs to transfer from a 400 km altitude circular orbit to a 600 km altitude circular orbit to intercept a piece of space junk. Calculate the total Δv required for this Hohmann transfer.

Given:

- Initial orbit: $h_1 = 400 \text{ km}$, so $r_1 = 6771 \text{ km}$
- Final orbit: $h_2 = 600 \text{ km}$, so $r_2 = 6971 \text{ km}$
- $\mu = 398600 \text{ km}^3/\text{s}^2$

ELI10 Concept: Imagine you're on a swing. To go higher, you need to push at just the right moment (at the bottom). In space, to change orbits, you fire your rocket engine at specific

points. A Hohmann transfer is like taking the most fuel-efficient path – it's like taking the "economy route" instead of the direct but expensive route!

Solution:

Step 1: Calculate velocities in the initial circular orbit

$$v_1 = \sqrt{\frac{\mu}{r_1}} = \sqrt{\frac{398600}{6771}} = 7.67 \text{ km/s} \quad (20)$$

Step 2: Calculate velocities in the final circular orbit

$$v_2 = \sqrt{\frac{\mu}{r_2}} = \sqrt{\frac{398600}{6971}} = 7.56 \text{ km/s} \quad (21)$$

Notice: The higher orbit is slower! This seems weird but remember – farther from Earth means weaker gravity pull, so less speed needed to stay in orbit.

Step 3: Calculate semi-major axis of the transfer ellipse

$$a_t = \frac{r_1 + r_2}{2} = \frac{6771 + 6971}{2} = 6871 \text{ km} \quad (22)$$

Step 4: Calculate velocity at periapsis (low point) of transfer orbit using vis-viva equation

$$v_{t1} = \sqrt{\mu \left(\frac{2}{r_1} - \frac{1}{a_t} \right)} = \sqrt{398600 \left(\frac{2}{6771} - \frac{1}{6871} \right)} = 7.73 \text{ km/s} \quad (23)$$

Step 5: Calculate velocity at apoapsis (high point) of transfer orbit

$$v_{t2} = \sqrt{\mu \left(\frac{2}{r_2} - \frac{1}{a_t} \right)} = \sqrt{398600 \left(\frac{2}{6971} - \frac{1}{6871} \right)} = 7.50 \text{ km/s} \quad (24)$$

Step 6: Calculate the two Δv burns

$$\Delta v_1 = v_{t1} - v_1 = 7.73 - 7.67 = 0.06 \text{ km/s} = 60 \text{ m/s} \quad (25)$$

$$\Delta v_2 = v_2 - v_{t2} = 7.56 - 7.50 = 0.06 \text{ km/s} = 60 \text{ m/s} \quad (26)$$

Step 7: Total Δv required

$$\Delta v_{total} = \Delta v_1 + \Delta v_2 = 60 + 60 = 120 \text{ m/s} \quad (27)$$

Answer

The total velocity change needed is **120 m/s**.

What this means: You fire your thrusters for a few seconds at the bottom of your orbit to speed up by 60 m/s, coast for about 45 minutes on the transfer ellipse, then fire again at the top to slow down by 60 m/s.

Fuel cost: Using the rocket equation with typical spacecraft ($v_e \approx 3000 \text{ m/s}$), this requires about 4% of your spacecraft's mass in propellant. That's why fuel is precious in space!

12.4 Example 4: Relative Motion - Catching Up to Debris (MEDIUM-HARD)

Problem

Your debris removal spacecraft is 100 m behind a tumbling satellite in the same circular orbit at 500 km altitude. If you apply a constant 0.1 N thrust forward for 60 seconds (spacecraft mass = 500 kg), what happens to your relative position?

Given:

- Initial separation: $y_0 = -100$ m (you're behind)
- Thrust: $F = 0.1$ N for $t = 60$ s
- Mass: $m = 500$ kg
- Orbit radius: $r = 6871$ km

ELI10 Concept: This is the weird part of orbital mechanics! In space, if you speed up to catch something in front of you, you actually end up going to a higher orbit and moving SLOWER, so you fall behind! It's like trying to run on a circular treadmill – pushing forward makes you move outward, not forward. This is why approaching space debris is so tricky!

Solution:

Step 1: Calculate orbital angular velocity n

$$n = \sqrt{\frac{\mu}{r^3}} = \sqrt{\frac{398600}{6871^3}} = 0.001115 \text{ rad/s} \quad (28)$$

Step 2: Calculate acceleration from thrust

$$a_y = \frac{F}{m} = \frac{0.1}{500} = 0.0002 \text{ m/s}^2 \quad (29)$$

Step 3: Calculate velocity change during burn

$$\Delta v_y = a_y \times t = 0.0002 \times 60 = 0.012 \text{ m/s} \quad (30)$$

Step 4: Use HCW equations for along-track motion For constant thrust in the along-track (y) direction, the solution is:

$$y(t) = y_0 + \frac{3na_y}{2n^2}(1 - \cos(nt)) + \frac{\Delta v_y}{n} \sin(nt) \quad (31)$$

After the 60 s burn stops, we have:

$$\dot{y}(60) = \Delta v_y = 0.012 \text{ m/s} \quad (32)$$

$$y(60) = -100 + (\text{small change during burn}) \approx -99.3 \text{ m} \quad (33)$$

Step 5: Predict long-term motion (after burn stops) The HCW equations show that with this velocity, you'll continue in a sinusoidal relative trajectory:

$$y(t) = -99.3 + \frac{0.012}{0.001115} \sin(0.001115 \times t) = -99.3 + 10.8 \sin(0.001115t) \quad (34)$$

Step 6: Key insight - radial coupling Because you thrust along-track, the HCW coupling term ($2n\dot{y}$) causes you to also move radially:

$$x(t) = \frac{2\Delta v_y}{n} = \frac{2 \times 0.012}{0.001115} = 21.5 \text{ m above the target} \quad (35)$$

Answer

Surprising result: After your burn:

- You move to an orbit **21.5 m higher** than the debris
- You oscillate forward and backward relative to it with amplitude ± 10.8 m
- You don't simply move forward – you create an elliptical relative trajectory!

What really happened: By pushing forward, you raised your orbit slightly. Higher orbits are slower, so now you drift backward! The relative motion creates a natural ellipse in the rotating frame.

Lesson for debris removal: To approach from behind, you paradoxically need to thrust *backward* to drop into a lower, faster orbit, then coast up to meet the target. Counter-intuitive, but that's orbital mechanics!

12.5 Example 5: Detumbling a Satellite (HARD)

Problem

A defunct satellite (500 mm cube, mass = 50 kg) is tumbling with angular velocity $\boldsymbol{\omega} = [0.1, 0.05, 0.15]$ rad/s. You want to reduce its tumbling to less than 0.01 rad/s in all axes using an ion beam that can apply 0.001 N·m of torque. How long will it take, and what's the strategy?

Given:

- Satellite: 0.5 m cube, mass $m = 50$ kg
- Initial angular velocity: $\boldsymbol{\omega}_0 = [0.1, 0.05, 0.15]$ rad/s
- Target: $|\boldsymbol{\omega}| < 0.01$ rad/s
- Available torque: $\tau_{max} = 0.001$ N·m

ELI10 Concept: Imagine a spinning ice skater. To stop spinning, they extend their arms (spreads out their mass). For a satellite, we can't change its shape, but we can apply a tiny "brake" force continuously. It's like gradually slowing down a spinning top by gently pressing your finger on it, except in space we use a beam of ions (charged particles) instead of your finger!

Solution:

Step 1: Calculate moment of inertia for a cube For a uniform cube with side length $L = 0.5$ m:

$$I = \frac{1}{6}mL^2 = \frac{1}{6} \times 50 \times 0.5^2 = 2.08 \text{ kg}\cdot\text{m}^2 \quad (36)$$

Assuming principal axes align with cube edges: $I_1 = I_2 = I_3 = I = 2.08 \text{ kg}\cdot\text{m}^2$

Step 2: Calculate initial angular momentum magnitude

$$|\mathbf{L}_0| = I|\boldsymbol{\omega}_0| = I\sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2} \quad (37)$$

$$= 2.08\sqrt{0.1^2 + 0.05^2 + 0.15^2} \quad (38)$$

$$= 2.08\sqrt{0.0325} = 2.08 \times 0.180 = 0.375 \text{ kg}\cdot\text{m}^2/\text{s} \quad (39)$$

Step 3: Calculate final angular momentum (target state)

$$|\mathbf{L}_f| = I \times 0.01 = 2.08 \times 0.01 = 0.0208 \text{ kg}\cdot\text{m}^2/\text{s} \quad (40)$$

Step 4: Angular momentum to be removed

$$\Delta L = |\mathbf{L}_0| - |\mathbf{L}_f| = 0.375 - 0.0208 = 0.354 \text{ kg}\cdot\text{m}^2/\text{s} \quad (41)$$

Step 5: Time required with constant torque Using $\tau = \Delta L / \Delta t$:

$$\Delta t = \frac{\Delta L}{\tau_{max}} = \frac{0.354}{0.001} = 354 \text{ seconds} \approx 6 \text{ minutes} \quad (42)$$

Step 6: Optimal detumbling strategy

The best approach is to apply torque opposite to the instantaneous angular momentum vector:

$$\boldsymbol{\tau}(t) = -\tau_{max} \frac{\boldsymbol{\omega}(t)}{|\boldsymbol{\omega}(t)|} \quad (43)$$

This means:

1. Measure current $\boldsymbol{\omega}$ using cameras/LIDAR
2. Point ion beam opposite to rotation direction
3. Update pointing every few seconds as rotation slows
4. Continue until all components < 0.01 rad/s

Step 7: Energy dissipated

$$E_0 = \frac{1}{2} I |\boldsymbol{\omega}_0|^2 = \frac{1}{2} \times 2.08 \times 0.180^2 = 0.034 \text{ J} \quad (44)$$

$$E_f = \frac{1}{2} I |\boldsymbol{\omega}_f|^2 = \frac{1}{2} \times 2.08 \times 0.01^2 = 0.0001 \text{ J} \quad (45)$$

$$\Delta E = 0.034 - 0.0001 \approx 0.034 \text{ J} \quad (46)$$

Answer

Time required: Approximately **6 minutes** of continuous ion beam operation.

Strategy:

1. Track satellite rotation using sensors (every 0.5 s)
2. Compute instantaneous spin axis direction
3. Point ion beam opposite to spin direction
4. Apply maximum torque (0.001 N·m)
5. Update beam direction as satellite slows
6. Stop when all rotation rates < 0.01 rad/s

Challenges:

- Must continuously track a tumbling object
- Ion beam pointing must be very accurate ($< 1^\circ$)
- Need to avoid creating new tumbling modes
- Must maintain safe distance (5-10 m) during operation

Real-world note: This is actually optimistic! Real debris might have unknown mass distribution, making I uncertain. Also, solar radiation pressure and gravity gradients can disturb the process. Actual detumbling missions might take 10-15 minutes.

12.6 Example 6: Complete ADR Mission Planning (VERY HARD)

Plan a complete Active Debris Removal (ADR) mission:

Initial State:

- Your chaser spacecraft: 400 km altitude circular orbit, inclination = 51.6°
- Debris (500 mm cube): 600 km altitude circular orbit, inclination = 51.6° , tumbling at 0.12 rad/s
- Chaser mass: 800 kg, fuel: 200 kg, $I_{sp} = 300$ s (chemical thruster)
- Time constraint: Complete mission in 48 hours

Mission phases:

1. Hohmann transfer to debris altitude
2. Approach to 50 m distance

3. Detumble debris
 4. Capture using robotic arm
 5. Deorbit to 250 km altitude (burns up in weeks)
- Calculate total Δv budget and verify mission feasibility.

ELI10 Concept: This is like planning a complete rescue mission in space! You need to: (1) drive your spaceship to the right highway (transfer orbits), (2) carefully approach the broken satellite (proximity operations), (3) stop it from spinning (detumbling), (4) grab it with a robot arm (capture), and (5) push it down toward Earth where it'll burn up safely (deorbit). Every step needs careful planning because you have limited fuel and time!

Solution:

PHASE 1: Hohmann Transfer (400 km $\rightarrow$ 600 km)

From Example 3, we know:

$$\Delta v_{transfer} = \Delta v_1 + \Delta v_2 = 120 \text{ m/s} \quad (47)$$

Transfer time (half orbital period of transfer ellipse):

$$a_t = \frac{6771 + 6971}{2} = 6871 \text{ km} \quad (48)$$

$$T_t = 2\pi \sqrt{\frac{a_t^3}{\mu}} = 2\pi \sqrt{\frac{6871^3}{398600}} = 5525 \text{ s} \quad (49)$$

$$t_{transfer} = \frac{T_t}{2} = 2763 \text{ s} \approx 46 \text{ minutes} \quad (50)$$

PHASE 2: Approach from 10 km to 50 m

Starting separation: 10 km along-track (phasing error after transfer)

Use V-bar approach (safer): Multiple small burns over several orbits

Small Δv to drop slightly below target: 2 m/s

Coast for 2-3 orbits to catch up: 3 hours

Small Δv to raise back to target: 2 m/s

Final approach burns (several iterations): 6 m/s

$$\Delta v_{approach} \approx 10 \text{ m/s}, \quad t_{approach} \approx 6 \text{ hours} \quad (51)$$

PHASE 3: Detumbling

From Example 5, with better thrusters:

$$t_{detumble} \approx 15 \text{ minutes} \quad (52)$$

Station-keeping during detumbling (maintaining 50 m distance):

$$\Delta v_{station} \approx 5 \text{ m/s}, \quad t_{detumble} = 15 \text{ min} \quad (53)$$

PHASE 4: Capture

Final approach (50 m $\rightarrow$ contact):

- Slow, controlled approach: 0.1 m/s approach velocity
- Multiple correction burns: 3 m/s total
- Time: 10 minutes

$$\Delta v_{capture} \approx 3 \text{ m/s}, \quad t_{capture} = 10 \text{ min} \quad (54)$$

PHASE 5: Deorbit (600 km $\rightarrow$ 250 km)

Combined mass: $m_{total} = 800 + 50 = 850 \text{ kg}$

Calculate Δv for deorbit burn:

$$r_1 = 6971 \text{ km (current orbit)} \quad (55)$$

$$r_2 = 6621 \text{ km (target perigee)} \quad (56)$$

$$a_d = \frac{r_1 + r_2}{2} = 6796 \text{ km} \quad (57)$$

Velocity at 600 km altitude:

$$v_1 = \sqrt{\frac{398600}{6971}} = 7.56 \text{ km/s} \quad (58)$$

Velocity needed for deorbit ellipse at periapsis:

$$v_d = \sqrt{398600 \left(\frac{2}{6971} - \frac{1}{6796} \right)} = 7.42 \text{ km/s} \quad (59)$$

$$\Delta v_{deorbit} = v_1 - v_d = 7.56 - 7.42 = 0.14 \text{ km/s} = 140 \text{ m/s} \quad (60)$$

TOTAL MISSION BUDGET

$$\Delta v_{total} = \Delta v_{transfer} + \Delta v_{approach} + \Delta v_{station} + \Delta v_{capture} + \Delta v_{deorbit} \quad (61)$$

$$= 120 + 10 + 5 + 3 + 140 \quad (62)$$

$$= 278 \text{ m/s} \quad (63)$$

Fuel Requirement Check

Using Tsiolkovsky equation:

$$\Delta v = v_e \ln \left(\frac{m_0}{m_f} \right) = g \cdot I_{sp} \cdot \ln \left(\frac{m_0}{m_f} \right) \quad (64)$$

where $v_e = g \cdot I_{sp} = 9.81 \times 300 = 2943 \text{ m/s}$

After capturing debris, total mass = 850 kg, including remaining fuel.

Required mass ratio:

$$\frac{m_0}{m_f} = e^{\Delta v/v_e} = e^{278/2943} = e^{0.0944} = 1.099 \quad (65)$$

Fuel needed for deorbit:

$$m_{fuel,deorbit} = m_{total}(1.099 - 1)/1.099 = 850 \times 0.090 = 76.5 \text{ kg} \quad (66)$$

For all mission phases (working backwards):

- Initial mass: 800 kg
- Fuel available: 200 kg
- Fuel for transfer + approach: $\sim 40 \text{ kg}$
- Remaining after capture: $\sim 160 \text{ kg}$
- Fuel needed for deorbit: 76.5 kg
- **Margin: $\sim 80 \text{ kg}$ (40%) $\checkmark$**

Timeline Check

$$t_{total} = t_{transfer} + t_{approach} + t_{detumble} + t_{capture} + t_{deorbit} \quad (67)$$

$$= 46 \text{ min} + 6 \text{ hr} + 15 \text{ min} + 10 \text{ min} + 5 \text{ min} \quad (68)$$

$$\approx 7.5 \text{ hours} \quad (69)$$

Final Answer - Mission Feasibility

Mission: FEASIBLE ✓

Summary:

- Total Δv required: **278 m/s**
- Fuel needed: **~120 kg** (60% of available)
- Safety margin: **40%** (good for uncertainties)
- Total mission time: **7.5 hours** (well within 48 hr limit)
- Debris disposal: Perigee at 250 km $\rightarrow$ natural decay in 2-4 weeks

Critical Success Factors:

1. Accurate orbit determination (< 100 m error)
2. Successful detumbling (otherwise capture fails)
3. Robust control during capture (free-floating base dynamics)
4. Contingency fuel reserve for emergencies (40% margin)

Risk Factors:

- Debris might fragment during capture (use gentle approach)
- Solar radiation pressure affects trajectory (~ 1 m/s over 48 hr)
- Sensor failures require backup navigation
- Combined system might be unstable after capture (need active control)

Why this works: The 40% fuel margin gives us room for:

- Unexpected maneuvers during approach
- Longer detumbling time if debris is worse than expected
- Multiple capture attempts if first one fails
- Emergency abort back to safe orbit

This is a realistic ADR mission profile similar to those being planned by ESA's ClearSpace-1 and other upcoming debris removal missions!

13 Key Equations Cheat Sheet

13.1 Orbital Mechanics

$$\text{Orbital velocity: } v = \sqrt{\frac{\mu}{r}} \quad (70)$$

$$\text{Orbital period: } T = 2\pi \sqrt{\frac{a^3}{\mu}} \quad (71)$$

$$\text{Vis-viva equation: } v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right) \quad (72)$$

$$\text{Hohmann } \Delta v : \Delta v_1 = \sqrt{\frac{\mu}{r_1}} \left(\sqrt{\frac{2r_2}{r_1 + r_2}} - 1 \right) \quad (73)$$

13.2 Relative Motion (HCW)

$$\ddot{x} - 2n\dot{y} - 3n^2x = \frac{f_x}{m} \quad (74)$$

$$\ddot{y} + 2n\dot{x} = \frac{f_y}{m} \quad (75)$$

$$\ddot{z} + n^2z = \frac{f_z}{m} \quad (76)$$

where $n = \sqrt{\mu/a^3}$

13.3 Attitude Dynamics

$$\mathbf{I} \cdot \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{I} \cdot \boldsymbol{\omega}) = \boldsymbol{\tau} \quad (77)$$

$$\text{Quaternion kinematics: } \dot{\mathbf{q}} = 0.5 \boldsymbol{\Omega}(\boldsymbol{\omega}) \mathbf{q} \quad (78)$$

13.4 Drag Force

$$F_{\text{drag}} = 0.5 \rho v^2 C_D A \quad (79)$$

where ρ = atmospheric density

13.5 Tsiolkovsky Rocket Equation

$$\Delta v = v_e \ln \left(\frac{m_0}{m_f} \right) \quad (80)$$

where v_e = exhaust velocity

14 Resources for Further Learning

14.1 Papers (from problem statement)

1. A. Ledkov, V. Aslanov, "Review of contact and contactless active space debris removal approaches", *Prog. Aerosp. Sci.* 134 (2022) 100858
2. M. Shan, J. Guo, E. Gill, "Review and comparison of active space debris capture and removal methods", *Prog. Aero. Sci.* 80 (2016) 18-32

14.2 Recommended Textbooks

- **Orbital Mechanics:** "Orbital Mechanics for Engineering Students" by Howard Curtis
- **Spacecraft Dynamics:** "Spacecraft Dynamics and Control" by Marcel Sidi
- **Space Robotics:** "Robotics and Automation in Space" – Yoshida/Umetani

14.3 Online Resources

- ESA Space Debris Office: https://www.esa.int/Space_Safety/Space_Debris
- NASA Orbital Debris Program: <https://orbitaldebris.jsc.nasa.gov/>

14.4 Software Tools

- **GMAT** (General Mission Analysis Tool): Free trajectory design
- **STK** (Systems Tool Kit): Commercial, orbit visualization
- **Orekit**: Open-source orbital mechanics library (Python/Java)

15 Summary: Your Robotics Skills Applied to Space

Terrestrial Robotics	Space Robotics (ADR)
Fixed base manipulator	Free-floating base
Cooperative targets	Non-cooperative, tumbling
Gravity assists	Orbital dynamics dominate
Unlimited power	Solar/battery constrained
Rich sensing	Limited, delayed sensing
Ground-based testing	Simulation + limited space testing
Reaction forces to ground	Momentum conservation

Your Advantage

Control theory, state estimation, path planning, and optimization translate directly! The physics is different but the mathematical frameworks are similar.

Good luck with your hackathon!